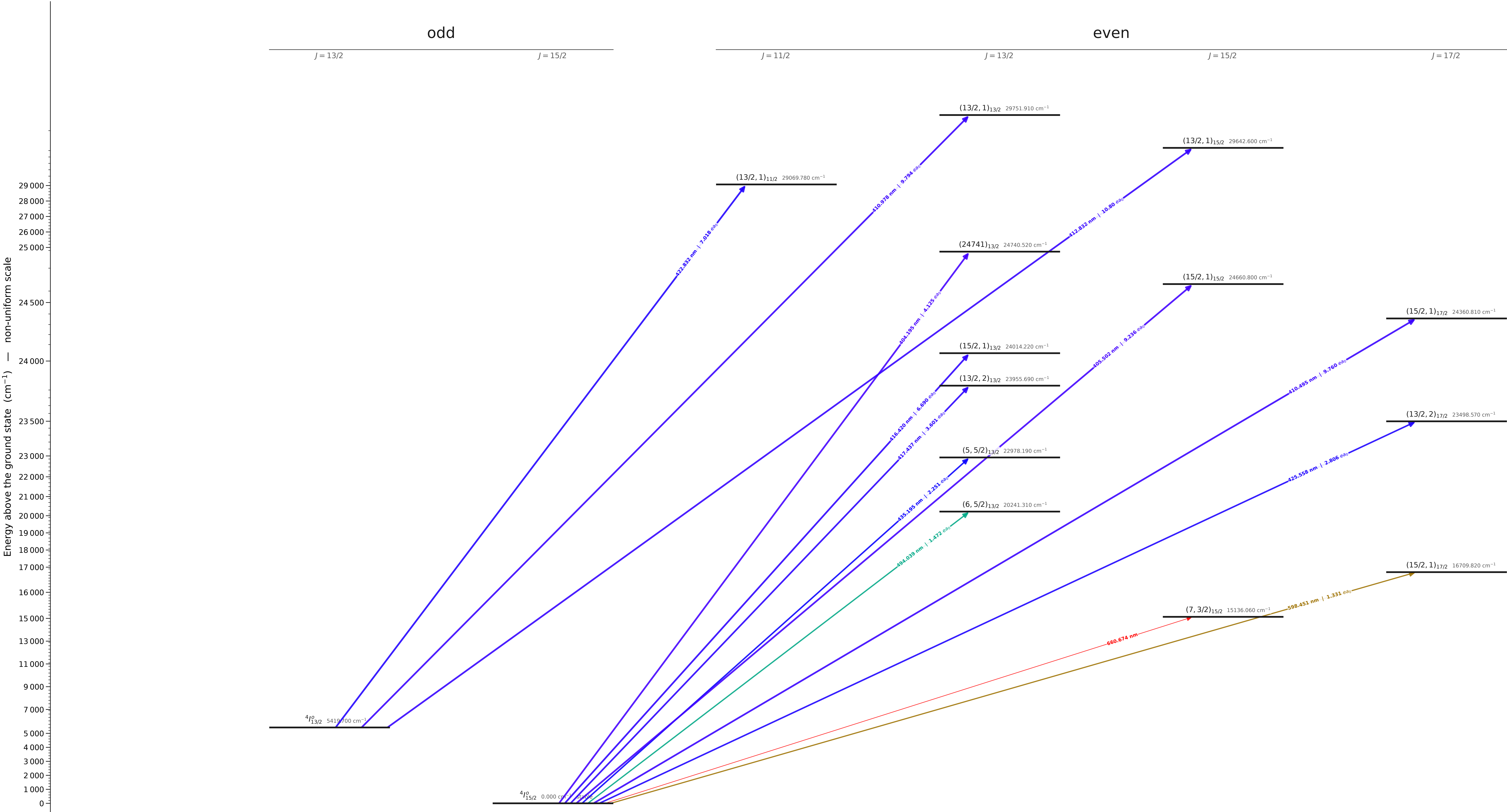
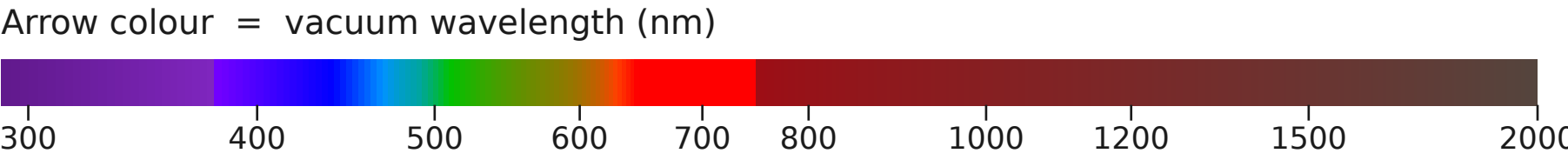


Holmium

Ho (Ho I). Strongest classified transitions below 2 μm (NIST): 15 levels, 13 lines.

Ho⁺ ionisation limit 48567.00 cm⁻¹ = 6.02154 eV (205.901 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle l || e r || l' \rangle|$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Label caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U[\text{erf}] \rangle$ (eao)	A (s ⁻¹)	f/2 π up. (MHz)	branch (%)	Use
$4^1_{15/2} - (24741, 1)_{13/2}$	404.1952	404.0811	741.7021	4.125	3.730e+07	-	-	-
$4^1_{5/2} - (15/2, 1)_{15/2}$	405.5018	405.3873	739.3122	9.236	1.620e+08	-	-	-
$4^1_{5/2} - (15/2, 1)_{11/2}$	410.4954	410.3796	730.3187	9.760	1.550e+08	-	-	-
$4^1_{13/2} - (13/2, 1)_{13/2}$	410.9779	410.8619	729.4613	9.794	2.000e+08	-	-	-
$4^1_{13/2} - (13/2, 1)_{15/2}$	412.8325	412.7161	726.1843	10.80	2.100e+08	-	-	-
$4^1_{15/2} - (15/2, 1)_{13/2}$	416.4199	416.3026	719.9282	6.690	8.970e+07	-	-	-
$4^1_{5/2} - (13/2, 2)_{13/2}$	417.4374	417.3197	718.1735	3.601	2.580e+07	-	-	-
$4^1_{5/2} - (13/2, 2)_{11/2}$	422.8316	422.7125	709.0116	7.018	1.100e+08	-	-	-
$4^1_{15/2} - (13/2, 2)_{11/2}$	425.5578	425.4380	704.4694	2.806	1.150e+07	-	-	-
$4^1_{5/2} - (5, 5/2)_{13/2}$	435.1953	435.0730	688.8688	2.251	8.900e+06	-	-	-
$4^1_{15/2} - (6, 5/2)_{13/2}$	494.0392	493.9013	606.8192	1.472	2.600e+06	-	-	-
$4^1_{5/2} - (15/2, 1)_{17/2}$	598.4505	598.2848	500.9478	1.331	9.300e+05	-	-	-

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.